

CO3 AT3

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I SRINIVASAN A 192411025 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A. Srinivasan

17+17+22+17+11
84

✓

CO3 - AT3 -
AI

Flight Delay Notification

1. Propositional Logic

Let

* D = Flight 202 is delayed

* N = Passengers are notified

Knowledge Base:

1. If a flight is delayed, then passengers are notified.

$$D \rightarrow N$$

2. Flight 202 is delayed

D

Goal:

N

Q. CNF and Negation of Goal

Convert implication:

$$D \rightarrow N \equiv \neg D \vee N$$

CNF clauses:

$$* C_1 = \neg D \vee N$$

$$* C_2 = D$$

Negate the goal:

$\neg N$

so add:

$$* C_3 = \neg N$$

Q. Resolution steps:

Resolve C_1 and C_2 :

$(\neg D \vee N), (D)$

Resolving D and $\neg D$:

N

Now resolve:

$(CN), (\neg N)$

Therefore



4. conclusion:

Since The Empty clause is derived:
Passengers are notified.

Q. Employee certification:

1. Propositional Logic

Let

T = Kevin completed training

C = Kevin receives certification

1. $T \rightarrow C$

2. T

Goal: C

2. $C \wedge \neg T$

$T \rightarrow C \equiv \neg T \vee C$

clauses:

$$*C_1 = \neg T \vee C$$

$$*C_2 = T$$

Negation of goal:

$$*C_3 = \neg C$$

3. Resolution

$$(\neg T \vee C), T \Rightarrow C$$

$$\therefore \neg C \Rightarrow \square$$

4. conclusion:

Kevin receives a certification

3. Medical Attention Alert.

1. Propositional Logic

Let:

F = Ravi has a fever

M = Ravi needs medical attention

$$1. F \rightarrow M$$

$$2. F$$

Goal: M

$$2. \neg M$$

$$F \rightarrow M \equiv \neg F \vee M$$

Clauses:

$$*C_1 = \neg F \vee M$$

$$*C_2 = F$$

Negation of goal:

$$\neg C_3 = \neg M$$

3. Resolution

$$(\neg F \vee M), F \Rightarrow M$$

$$M, \neg M \Rightarrow \square$$

4. conclusion:

Ravi needs medical attention.

4. Library Late fine.

1. Propositional Logic.

Let:

L = Book was returned late

F = A fine is charged

$$1. L \rightarrow F$$

$$2. L$$

Goal: F

2. CNF

$$L \rightarrow F \equiv \neg L \vee F$$

clauses:

$$\neg C_1 = \neg L \vee F$$

$$\neg C_2 = L$$

Negation of goal:

$$\neg C_3 = \neg F$$

Resolution

$$(L \vee F), L \Rightarrow F$$

$$F, \neg F \Rightarrow \square$$

4. conclusion

A fine is charged.

5. Order shipping confirmation.

Let:

P = Payment is verified

C = order is confirmed

S = item is shipped

$$1. P \rightarrow C$$

$$2. C \rightarrow S$$

$$3. P$$

Goal: S

2. CNF

$$P \rightarrow C \equiv \neg P \vee C$$

$$C \rightarrow S \equiv \neg C \vee S$$

Clauses:

$$*C_1 = \neg P \vee C$$

$$*C_2 = \neg C \vee S$$

$$*C_3 = P$$

3. Negation of Goal

$$C_4 = \neg S$$

4. Resolution steps :

Step 1:

$$(\neg P \vee C), P \Rightarrow C$$

Step 2:

$$(\neg C \vee S), C \Rightarrow S$$

Step 3:

$$S, \neg S \Rightarrow \boxed{\square}$$

∴ conclusion :

Since the empty clause is derived,

~~The item~~ is shipped.